Linear-Quadratic Systems

Name _____

Algebra 2 Lesson 2-7 (fast track) Date _____ Period _____

Learning goal: I can determine how many real solutions a linear-quadratic system has and solve a quadratic equation by graphing a linear-quadratic system and finding the intersection points.

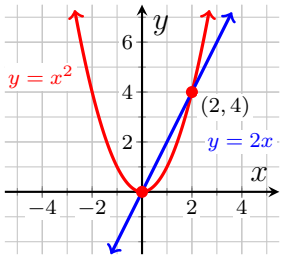
Language goal: I can explain how many solutions a system has using the words system, intersection, tangent, and approximate.

On the quiz: (1) Count the real solutions of a line and a parabola (Example 1, Try It 1). (2) Bonus: solve a system whose two equations both start "$y =$" (the Example 1 check; Exit question). (3) A ball up a hill and (4) shading inequalities: full Savvas quiz only. (5) Solve with a calculator's intersection feature (Example 5, Try It 5).

On the Topic 2 test: Question 16 asks how many real solutions a parabola and a line have (Example 1). Question 17 asks you to solve a quadratic equation with a calculator's intersection feature, to the nearest hundredth (Example 5).

Example 1: Determine the Number of Solutions

How many solutions can there be for a linear-quadratic system?



A. How many real solutions does the **system** $y = x^2$, $y = 2x$ have?

The graph shows the parabola and the line meet at two points, $(0, 0)$ and $(2, 4)$. A solution is an ordered pair that makes *every* equation in the system true, so the solutions are the points of **intersection**.

Check with algebra: both equations give y, so set them equal: $x^2 = 2x$, $x^2 - 2x = 0$, $x(x - 2) = 0$, $x = 0$ or $x = 2$.

This system has two real solutions.

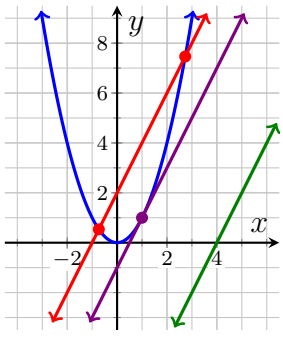
B. How many solutions could this system have? $y = x^2$, $y = 2x + b$

Test values for b to determine when the system has different numbers of solutions.

- $b = 2$: two solutions; the graphs intersect at two points.
- $b = -1$: one solution; $y = 2x - 1$ is **tangent** to $y = x^2$ at $(1, 1)$.
- $b = -8$: no solutions; $y = 2x - 8$ does not intersect $y = x^2$.

There is no way for a line to cross a parabola more than twice.

A linear-quadratic system may have 0, 1, or 2 solutions.

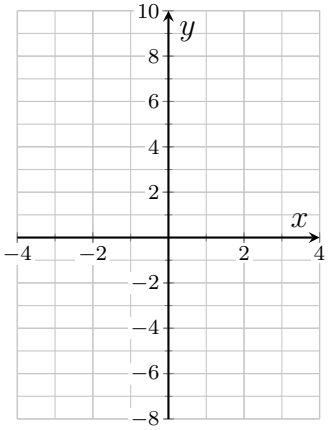


$y = x^2$; $y = 2x + 2$, $y = 2x - 1$, $y = 2x - 8$

Say it: The line $y = 2x - 1$ is _____ to the parabola, so the _____ has _____ solution(s). A line and a parabola can have 0, 1, or 2 points of _____ because ...

Try It 1: Count the solutions

Determine the number of real solutions of the system $y = 3x^2$, $y = 3x - 2$.
Hint: graph both on the grid, or set $3x^2 = 3x - 2$ and check the discriminant (Lesson 2-6).



Number of real solutions: _____

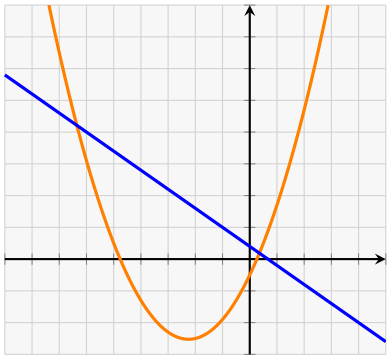
Example 5: Using a System to Solve an Equation

Solve the equation $x^2+9x-5 = 4-3x$ by writing a linear-quadratic system and using the intersection feature of a graphing calculator to solve it.

1. Each side of the equation becomes one equation of the **system**:

$$y = x^2 + 9x - 5, \qquad y = 4 - 3x$$

2. Graph the system. The graphing calculator shows the curve and line **intersect** at $x \approx 0.71$ and $x \approx -12.71$. These are **approximate** solutions.
Hint: on a TI-84, CALC then 5:intersect, once for each crossing; in Desmos, click each point where the graphs cross.



Xscl: 2 Yscl: 10

3. Check: substitute into the original equation; one side should be approximately equal to the other.

$$(0.71)^2 + 9(0.71) - 5 \stackrel{?}{=} 4 - 3(0.71)$$
$$0.5041 + 6.39 - 5 \stackrel{?}{=} 4 - 2.13$$
$$1.8941 \approx 1.87 \checkmark$$

$$(-12.71)^2 + 9(-12.71) - 5 \stackrel{?}{=} 4 - 3(-12.71)$$
$$161.5441 - 114.39 - 5 \stackrel{?}{=} 4 + 38.13$$
$$42.1541 \approx 42.13 \checkmark$$

The solutions are $x \approx 0.71$ and $x \approx -12.71$.

Say it: I wrote the _____ $y = x^2 + 9x - 5$, $y = 4 - 3x$. The x-values of the _____ points are the solutions. They are _____, so the two sides of the check are close but not equal.

Try It 5: Solve with the intersection feature

Solve the equation $3x^2 - 7x + 4 = 9 - 2x$ by writing a linear-quadratic system and solving using the intersection feature of a graphing calculator.

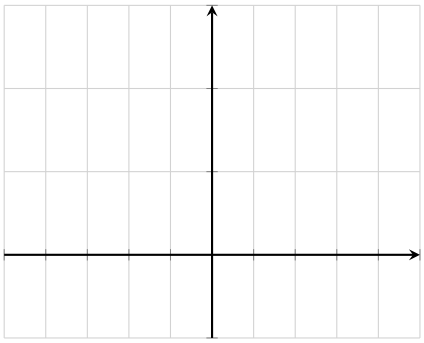
System: $y =$ _____
$y =$ _____

The curve and line intersect at

$x \approx$ _____ and $x \approx$ _____

Check one value in the original equation:

Sketch what the screen shows. Mark each **intersection**.



IXL: Practice for the quiz

Go to IXL, Algebra 2. Type the code in the search box. Do the starred skills first.

Code	Skill	Feeds
VQU ★	Solve a linear-quadratic system of equations by graphing: lines and parabolas	Quiz 1: count where the line meets the parabola (Example 1); Quiz 5: read the intersections (Example 5)
HVZ ★	Solve a linear-quadratic system of equations: lines and parabolas	Quiz 2 (bonus): set the two y-expressions equal and solve (the Example 1 check)
BW5	Solve a system of equations using substitution	Warm-up: the same move with two lines (Lesson 1-6)
M69	Solve a system of equations by graphing	Warm-up: read where two graphs meet (Lesson 1-5)

SmartScore 70 or above on each starred skill replaces your quiz score.

Exit question: Substitute and solve

Use substitution to solve the system of equations.

$$\begin{cases} y = -x^2 + x + 12 \\ y = 2x - 8 \end{cases}$$

Solutions: _____ and _____

Extension (self-paced, DOK 3, Practice 29): A golfer accidentally hits a ball toward a water hazard that is downhill from current position on the fairway. The hill can be modeled by a line through the origin with slope $-\frac{1}{8}$. The path of the ball can be modeled by the function $y = -\frac{1}{100}x^2 + \frac{3}{2}x$. **Part A** If the golfer stands at the origin, and the water hazard is 180 yd away, will the golfer's ball bounce or splash? **Part B** How far did the ball land from the edge of the water hazard? **Part C** Does it matter whether you measure the 180 yd horizontally or along the hill? Explain.

Before the quiz, finish these sentences.

The solutions of a _____ are the points of _____ of its graphs.

A line that is _____ to a parabola touches it once, so that system has one solution; a line and a parabola can have 0, 1, or 2.

A calculator gives _____ solutions, so I check them in the original equation and the two sides are close.